Felipe Nunes

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Embedded Systems Engineer

Education

University of British Columbia — Sep 2020 – May 2026 (Expected)
Bachelor of Applied Science in Integrated Engineering

Professional Experience

Miru Smart Technologies — May 2024 – Aug 2024
Systems Integration Intern — *Vancouver, B.C.*

- Reduced PCB footprint by 60% (6-layer, BGA SoM) and added a tuned 2.4 GHz inverted-F antenna
- Wrote C modules in Zephyr to drive electrochromic windows controller, increasing transitioning rate by 50%
- Devised a custom compilation pipeline using CMake and linker scripts for an ARM64 target

Sarcomere Dynamics — May 2023 – Apr 2024
Electrical and Firmware Engineering Intern — *Vancouver, B.C.*

- Implemented embedded C firmware with a finite state machine to control actuators in a multi-DOF robotic hand
- Redesigned three PCBs with emphasis on USB-C integration, mixed-signal routing, and power delivery
- Developed modular C++ classes and extended a custom Python API to support wireless communication via Wi-Fi
- Performed reflow soldering, signal probing, and hardware bring-up for embedded prototypes

Projects

Robotic Chessboard | Code | Report — Sep 2024 – Apr 2025

- Won the Best Engineering Project award out of all final-year Capstone projects
- Designed custom PCBs and cable assemblies for a multiplexed Hall effect sensor grid to detect chess piece positions
- Selected stepper motors and drivers; configured and compiled a custom GRBL build for an H-bot gantry
- Developed hardware abstraction layers and implemented a backend algorithm for sensor polling and state detection

Mountable Heads Up Display | Code | Report — Oct 2022 – Apr 2023

- Engineered an ESP32-based control unit to interface with UBC Formula Electric's CAN bus
- Displayed real-time telemetry (speed, temp, lap time) on an I2C OLED using FreeRTOS task scheduling
- Designed a compact custom PCB integrated into a 3D-printed helmet-mounted enclosure

Automated Drink Dispenser | Code | Report — Nov 2021 – Apr 2022

- Constructed a custom beverage dispenser using CNC-milled aluminum and a 3D printed nozzle
- Controlled dosing system with solenoid valves and load cell feedback for precision pouring
- Wrote firmware for automated calibration, stepper control, drink selection and dispensing, and LCD user interface

Engineering Design Teams

Open Robotics — Sep 2024 – Present
Grippers Team Lead — *University of British Columbia*

- Directed the full-cycle development of a parallel gripper end effector for the RoboCup@Home competition
- Managed a cross-disciplinary team of eight from initial design and prototyping to system integration

Supermileage — Sep 2022 – Aug 2023
Electrical Team Member — *University of British Columbia*

- Redesigned and assembled a custom PCB leveraging analog sensors to detect faults in a hydrogen electric FSAE car
- Competed and won 2023 SEMA competition in the hydrogen fuel cell category at the Indianapolis Motor Speedway

Technical Skills

Operating Systems: Linux (Arch, Ubuntu, Raspbian), Windows, macOS, FreeRTOS, Zephyr
Programming Languages: C/C++/C#, Python, JavaScript, HTML, CSS, MATLAB, R, Shell Scripting, Make, CMake
Hardware Design: Circuit design/analysis, PCB layout (Altium, KiCad), FPGAs, VHDL, Multisim, SPICE
Mechanical Design: Fusion 360, Onshape, SolidWorks, Orca Slicer
Tools & Instrumentation: Oscilloscopes, Logic Analyzers, Function Generators, Soldering (reflow & hand)

Languages

French (DELF B2), Portuguese (Bilingual Proficiency), Spanish (Working Proficiency)